

RAD (Rapid Application Development) is a software development methodology that emphasizes **quick development and iteration** of prototypes over strict planning and long development cycles. It's ideal for projects where requirements evolve through user feedback and where fast delivery is important.

Key Characteristics of RAD:

- Focus on **prototypes** instead of extensive documentation
- **User involvement** throughout the development process
- **Fast development cycles** with frequent iterations
- **Reusable components** to reduce effort
- Best suited for **UI/UX-intensive projects** like web and mobile apps

Phase 1: Requirements Planning

This is where the **project goals, system requirements, and user needs** are identified. Stakeholders, users, and designers collaborate to decide what needs to be built.

1) Identify Key Features

Navigation:

- Home
- Product Details
- Product Categories
- Order Confirmation

User Actions: Browsing, Searching, Adding to Cart, Checkout, Tracking Orders

2) Create a Requirements Document:

1. View list of product categories
2. View products under each category
3. View detailed information about a product
4. View and edit cart
5. Checkout and enter user/shipping/payment details
6. View order confirmation message

Phase 2: User Design

Designers and users collaborate to **create wireframes and prototypes**. The idea is to show how the application will look and behave before any actual coding starts.

1) Tools Used: Axure: Used to design the wireframes and UI prototype.

2) Wireframes Designed:

- **Home Page:** Highlights featured products and links to categories.
- **Product Categories:** Lists all product categories to explore.
- **Product Listings:** Shows products under a selected category.
- **Product Details:** Displays product info with "Add to Cart" option.
- **Cart:** Shows added items with price and quantity details.
- **Checkout:** Collects shipping and payment information.
- **Order Confirmation:** Confirms successful order placement.

Designed **interactions and navigation** using Axure's Prototype mode Reused components like headers/footers using **Components** Annotated elements for clarity (e.g., what a button does, purpose of sections)

Phase 3: Construction

This is where the **actual development happens**, but in RAD, it is tightly coupled with ongoing testing and feedback. You refine the prototype based on user responses.

1) Interactive Prototypes:

- Created using Axure's **prototype linking** features.
- Functional paths designed:
 - From **Home** → **Product Categories** → **Product Listings** → **Product Details** → **Add to Cart** → **Checkout** → **Order Confirmation**

2) Dynamic Elements:

- Simulated transitions and modals for adding to cart and checking out.
- Screens simulate real-time behavior such as navigating between product listings and cart.

3) Testing and Iteration:

- Prototype previewed with **"Present"** mode in Figma.
- Collected feedback from stakeholders (assumed for documentation).
- Iterated based on visual hierarchy, accessibility, and flow suggestions.

Phase 4: Cutover

In this final phase, the system goes live. This includes **finalizing the design, training users, and preparing documentation** for developers.

1) Finalization and Export:

- Final design prototype shared through Figma link.
- Ready for hand-off to development team.

- Could be exported to **HTML/CSS** using plugins like Figma-to-HTML or further tools if needed.

2) User Training & Support:

- Walkthrough of the prototype for stakeholders or development team. • Notes/documentation provided for developers to understand functionality and user flow.
- Easy access to Figma comments for further collaboration and questions.



